Isogeny-class rho at 46 bits (replica of P1)

Classification: replication of a null result (rho against rho on one instrument); not an index-calculus improvement, no leaderboard change, no claim about any deployed curve.

Protocol: `PROTOCOL.md` (frozen before data, sha256 `f1a2fbd5…6287`), decision rules R1–R6 inherited from `research/isogeny-class-rho/`. Run: `runs/run-20261019-n45/` (command card `run.sh`, outputs under `P2/`). Tool: `src/bin/isoclass_rho` (merged in #2174).

Replica at 46 bits: the null holds at 1,773× the group order. 4,400 class-mode rho solves on 38 isogenous curves of one prime-field class (P2, n = 46,339,405,405,021), every solve returning the planted k. Pooled mean 1.0097 × √(πn/2) (bootstrap 95% CI 0.993 to 1.026 of √(πn/2)); Kruskal–Wallis p = 0.38, ANOVA p = 0.67, no curve flagged (max |z| 2.39 against Bonferroni 3.21). The mean ratio P2/P1 is 41.68 against √(n_P2/n_P1) = 42.11 (0.990). Replay reproduces 220/220 iteration counts exactly. **One deviation:** the generators family (20 generators × 2 vertices) failed the frozen rule (KW p = 1.2e-4, two arms flagged); an unplanned fresh-seed repeat written down before running (`ADDENDUM-generators.md`) did **not** replicate it (KW p = 0.97). Both are reported; the first is unexplained.

What was measured

One class, P2: `F_p`, p = 46339417407469, a = 4184642753251, b = 42214793703111, `#E = n` = 46,339,405,405,021 (46 bits, prime order). The walk carries one ECDLP instance by verified isogenies; rho (32-adding walk, distinguished points, 8 bits) is run on 38 isogenous curves × 100 seeds, plus 3 isomorphism/representation controls, an A/A replica, a planted negation fold and a planted 4-adding walk.

item	value	expected
solves in class mode, all correct	4,400	
pooled mean / √(πn/2)	1.0097	1.016 (32-adding)
Kruskal–Wallis / ANOVA across curves	p 0.38 / 0.67	
flagged curves	0 of 38	
controls iso0, iso1, iso2, A/A	1.107, 0.996, 0.984, 0.974	1
planted negation fold	0.732	0.707
planted 4-adding walk	1.291	1.137 (Teske); P1 read 1.27 to 1.36
replay, first 5 seeds	220 of 220 iteration counts identical	R6
mean ratio P2/P1	41.68	√(n_P2/n_P1) = 42.11

Generators family and its repeat

run	seed	solves	KW p	ANOVA p	variance ratio	max abs z (crit 3.23)	flagged
generators (frozen card)	20261018	400	1.2e-4	2.7e-5	2.34	4.76	v158g12 (1.64×), v158g17 (0.61×)
gen-rep1 (`ADDENDUM-generators.md`, sha256 `571e9361…f17baf`)	20261019	400	0.97	0.98	0.58	1.97	none

By the frozen verdict wording the first is "more extreme than the family". The repeat is a different draw of generators and not a preregistered confirmation; it did not reproduce the effect and had under-dispersed arm means. This does not explain the first result. The first study had one flagged generator in 384 (B1); the same family type here has two in 40. Treat generator-level heterogeneity as unresolved, with more generator draws as the next measurement.

Limits

- One class at one larger size, one seed set; power about 18 % per curve, 2 % class-wide at 100 seeds. A null does not exclude smaller effects.
- Same shared VM as the first run (Xeon @ 2.10 GHz, 4 threads, isolation L0). Wall-time columns are unisolated observations, not admitted performance evidence.
- The reviewer-noted native KS tie defect and shared-seed covariance concerns on the first study apply to this tool too; the Kruskal–Wallis and ANOVA values above do not depend on the KS routine.
- No ecbench spec/session/audit records exist for this run (AGENTS.md §12); it cannot support a comparison claim.

Results

The four classes

class	field	root curve	`#E = h·n`	t	`D = f²·D_K`	`h(D_K)`	class size	strata walked / measured	walked / measured curves	edges verified
P2	`F_p`, p = 46339417407469	a = 4184642753251, b = 42214793703111 (j = 11686938476036)	1 · 46339405405021 (46 bits)	12002449	$315^2 \cdot (-416214539)$, f = $3^2 \cdot 5 \cdot 7$	8,332	2,624,580	10 / 12	469 / 38	53

Class size is $\Sigma_{d \mid f} h(d^2 D_K)$ with $h(D_K)$ counted from reduced forms and the other class numbers from the conductor formula; $h(f^2 D_K)$ was also counted directly and agrees in every class (`h_full_order_counted` = `h_full_order_formula` in each `walk.json`).

The isogenies walked, easy and hard

class	ℓ	role	volcano depth, crater	route	kernel field degree d	tree edges realised (neighbour searches)	mean ms per kernel search
P2	3	everywhere	2, split	elkies	1 (eigen-orbit 1)	7 (1644)	1.0
P2	5	everywhere	1, split	elkies	2 (eigen-orbit 2)	16 (814)	0.9
P2	7	everywhere	1, split	elkies	3 (eigen-orbit 3)	18 (796)	1.2
P2	13	everywhere	0, ramified	elkies	6 (eigen-orbit 6)	4 (107)	1.2
P2	23	hard (first vertices only)	0, split	elkies	11 (eigen-orbit 11)	3 (6)	50.2

P2	41	hard (first vertices only)	0, split	elkies	2, 10 (eigen-orbit 10)	2 (4)	913.1
P2	47	hard (first vertices only)	0, split	elkies	23 (eigen-orbit 23)	1 (4)	1,794.8
P2	59	hard (first vertices only)	0, split	elkies	29 (eigen-orbit 29)	2 (5)	4,212.0
P2	2	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	11	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	17	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	19	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	29	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	31	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	37	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	43	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	53	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			
P2	61	inactive	0, inert	inert and $\ell \mid f$: no rational ℓ-isogeny			

P2: every measured curve, one instance, 100 paired seeds per arm

Reference $\sqrt{(\pi n/2)}$ = 8,531,692 (×1.016 for the 32-adding walk: 8,668,208); 4400 solves, all returned the planted k; rho wall 4,036.4 s on 4 threads.

arm	kind	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	mean ms	e2e ms	ratio to $\sqrt{(\pi n/2)}$	z vs rest
all isogenous	pooled	38	3800	124,133	8,614,326 ± 143,001	8,118,256	32,970,081	0.522	3,664.0		1.010	
controls	iso / representation	3	300	311,784	8,778,216 ± 491,454	8,462,148	23,412,682	0.495	3,723.7		1.029	
iso0	iso-control	1	100	311,784	9,443,666 ± 908,618	9,336,226	20,569,743	0.491	4,009.1	4,009.1	1.107 [1.009, 1.214]	1.77
iso1	iso-control	1	100	1,184,106	8,493,398 ± 855,326	7,952,922	23,412,682	0.514	3,612.5	3,612.5	0.996 [0.902, 1.099]	−0.27
iso2	iso-control	1	100	415,793	8,397,585 ± 777,985	8,213,201	18,729,450	0.473	3,549.5	3,549.5	0.984 [0.899, 1.080]	−0.54
aa	aa-replica	1	100	850,318	8,313,456 ± 714,366	8,635,492	18,285,782	0.438	3,586.1	3,586.1	0.974 [0.896, 1.062]	−0.81
plant-neg	plant (expected 0.707)	1	100	211,921	6,244,080 ± 731,949	6,094,106	21,785,029	0.598	2,774.2	2,774.2	0.732 [0.651, 0.823]	−6.23
plant-r4	plant (expected 1.137)	1	100	471,364	11,018,469 ± 1,096,548	10,286,566	24,124,986	0.508	4,663.3	4,663.3	1.291 [1.173, 1.422]	4.26

Pooled by endomorphism-ring stratum (conductor of End relative to the walked primes), by level in each volcano, by distance from the root, by the degree and kernel field of the edge that reached the curve:

group	curves	solves	min	mean ± 95 % CI	median	max	sd / mean	ratio to $\sqrt{(\pi n/2)}$	z vs rest
conductor:1	1	100	962,643	8,762,579 ± 804,022	8,434,716	19,007,148	0.468	1.027	0.37
conductor:105	7	700	318,590	8,596,690 ± 342,549	7,837,885	32,970,081	0.538	1.008	−0.11
conductor:15	3	300	258,501	8,817,782 ± 494,522	8,547,081	23,626,980	0.496	1.034	0.84
conductor:21	3	300	518,996	8,116,111 ± 503,308	7,490,892	25,150,503	0.548	0.951	−2.02
conductor:3	3	300	266,288	8,590,737 ± 512,312	8,521,565	22,595,695	0.527	1.007	−0.09
conductor:315	3	300	570,978	8,594,920 ± 503,404	8,188,677	22,240,447	0.518	1.007	−0.08
conductor:35	3	300	270,157	8,307,934 ± 500,341	7,632,532	24,197,893	0.532	0.974	−1.25
conductor:45	3	300	124,133	8,601,602 ± 509,890	8,130,973	26,293,936	0.524	1.008	−0.05
conductor:5	3	300	517,281	8,786,835 ± 500,439	8,380,401	22,178,203	0.503	1.030	0.70
conductor:63	3	300	153,958	8,863,114 ± 506,783	8,398,579	24,878,026	0.505	1.039	1.00
conductor:7	3	300	858,106	8,786,666 ± 518,912	8,456,013	23,827,598	0.522	1.030	0.68
conductor:9	3	300	224,426	8,669,300 ± 521,924	8,252,741	23,716,356	0.532	1.016	0.22
dist:0	1	100	1,208,721	8,719,464 ± 843,204	8,109,277	20,506,203	0.493	1.022	0.25
dist:1	8	800	518,996	8,623,455 ± 318,912	8,086,108	32,970,081	0.534	1.011	0.06
dist:2	6	600	424,340	8,542,611 ± 360,306	8,128,124	23,827,598	0.527	1.001	−0.43
dist:3	23	2300	124,133	8,625,289 ± 182,693	8,123,997	26,293,936	0.518	1.011	0.19
ell3:level0	10	1000	270,157	8,640,688 ± 275,194	8,115,243	24,197,893	0.514	1.013	0.22
ell3:level1	16	1600	258,501	8,546,920 ± 221,855	8,027,050	32,970,081	0.530	1.002	−0.79
ell3:level2	12	1200	124,133	8,682,234 ± 255,028	8,231,478	26,293,936	0.519	1.018	0.63
ell5:level0	16	1600	153,958	8,615,023 ± 220,771	8,252,741	25,150,503	0.523	1.010	0.01
ell5:level1	22	2200	124,133	8,613,820 ± 187,740	8,047,607	32,970,081	0.522	1.010	−0.01

ell7:level0	16	1600	124,133	8,697,584 ± 218,507	8,346,504	26,293,936	0.513	1.019	0.98
ell7:level1	22	2200	153,958	8,553,775 ± 189,095	7,991,823	32,970,081	0.529	1.003	−0.98
field_degree:1	3	300	570,978	8,460,583 ± 504,558	7,978,601	22,762,702	0.527	0.992	−0.62
field_degree:10	1	100	590,366	8,750,325 ± 888,182	7,837,885	21,864,753	0.518	1.026	0.30
field_degree:11	2	200	1,044,901	8,563,652 ± 649,854	8,086,108	25,137,985	0.548	1.004	−0.16
field_degree:2	11	1100	153,958	8,644,006 ± 271,789	8,035,300	32,970,081	0.532	1.013	0.26
field_degree:3	16	1600	124,133	8,697,584 ± 218,507	8,346,504	26,293,936	0.513	1.019	0.98
field_degree:6	4	400	270,157	8,280,037 ± 426,320	7,745,654	24,197,893	0.525	0.971	−1.62
parent_ell:13	4	400	270,157	8,280,037 ± 426,320	7,745,654	24,197,893	0.525	0.971	−1.62
parent_ell:23	2	200	1,044,901	8,563,652 ± 649,854	8,086,108	25,137,985	0.548	1.004	−0.16
parent_ell:3	3	300	570,978	8,460,583 ± 504,558	7,978,601	22,762,702	0.527	0.992	−0.62
parent_ell:41	2	200	590,366	8,620,820 ± 653,992	7,824,010	32,970,081	0.547	1.010	0.02
parent_ell:5	10	1000	153,958	8,659,275 ± 283,145	8,076,650	25,150,503	0.528	1.015	0.36
parent_ell:7	16	1600	124,133	8,697,584 ± 218,507	8,346,504	26,293,936	0.513	1.019	0.98

Across the 38 measured curves: Kruskal–Wallis H = 39.08 (df 37, p = 0.377); ANOVA F = 0.882 (p = 0.674); between-curve variance ratio 0.882 (observed sd of curve means 422,588 against 449,754 expected from the pooled single-solve spread); largest |z| 2.39 against the Bonferroni threshold 3.21; curves beyond it: none.

Per curve (iterations = group operations; `d` = kernel field degree of the edge that reached it; transport = ms to find and apply the isogenies from the root):

curve	stratum f	levels	path (ℓ)	d	transport ms	min	mean ± CI	median	max	p10	p90	sd/mean	skew	ns/op	e2e ms	ratio	z
v218	1	3:0,5:0,7:0	3·5·7	3	3.4	962,643	8,762,579 ± 804,022	8,434,716	19,007,148	3,748,323	14,144,903	0.468	0.40	423.1	3,711.4	1.027	0.37
v52	3	3:1,5:0,7:0	5·7	3	1.4	1,377,977	8,490,494 ± 850,566	8,725,589	20,463,878	3,064,513	13,882,837	0.511	0.56	423.9	3,597.2	0.995	−0.29
v254	3	3:1,5:0,7:0	5·7·7	3	6.7	266,288	8,575,391 ± 903,791	8,662,256	19,965,117	3,163,324	14,319,000	0.538	0.55	423.9	3,643.3	1.005	−0.09
v363	3	3:1,5:0,7:0	41·5·7	3	31.7	576,611	8,706,326 ± 914,801	8,497,198	22,595,695	2,695,668	14,927,592	0.536	0.36	422.8	3,717.8	1.020	0.20
v45	5	3:0,5:1,7:0	3·7	3	2.8	890,559	8,433,792 ± 908,621	7,980,204	21,849,340	2,878,618	15,309,551	0.550	0.72	427.1	3,609.2	0.989	−0.39
v225	5	3:0,5:1,7:0	3·7·7	3	8.1	1,576,576	8,261,884 ± 787,398	8,160,413	22,099,292	3,151,440	13,401,582	0.486	0.56	425.1	3,520.0	0.968	−0.89
v353	5	3:0,5:1,7:0	41·3·7	3	32.7	517,281	9,664,828 ± 882,102	9,622,994	22,178,203	4,174,205	15,256,278	0.466	0.44	426.1	4,138.4	1.133	2.37
v44	7	3:0,5:0,7:1	3·5	2	2.5	1,008,007	8,886,516 ± 981,714	8,652,227	23,827,598	2,597,742	15,133,380	0.564	0.64	427.7	3,796.6	1.042	0.55
v215	7	3:0,5:0,7:1	3·5·5	2	5.7	878,780	8,685,856 ± 924,324	8,101,166	23,151,699	2,920,147	15,223,846	0.543	0.53	424.8	3,699.1	1.018	0.15
v352	7	3:0,5:0,7:1	41·3·5	2	32.5	858,106	8,787,625 ± 787,926	8,638,860	17,787,557	3,488,654	14,306,413	0.457	−0.00	424.1	3,760.2	1.030	0.44
v107	9	3:2,5:0,7:0	3·5·7	3	3.3	224,426	8,231,188 ± 942,481	7,762,692	21,651,223	2,872,408	15,820,182	0.584	0.61	427.1	3,511.7	0.965	−0.81
v146	9	3:2,5:0,7:0	3·5·7	3	3.1	666,256	9,321,601 ± 902,718	9,260,902	19,575,416	3,577,003	15,846,076	0.494	0.30	424.6	3,962.2	1.093	1.56
v185	9	3:2,5:0,7:0	3·5·7	3	3.5	439,305	8,455,110 ± 859,791	7,879,701	23,716,356	3,403,636	13,705,757	0.519	0.76	424.5	3,595.9	0.991	−0.37
v6	15	3:1,5:1,7:0	7	3	0.8	970,152	9,285,022 ± 914,332	9,367,019	23,626,980	3,793,833	14,836,504	0.502	0.45	426.4	3,948.4	1.088	1.46
v244	15	3:1,5:1,7:0	5·5·7	3	4.1	258,501	8,691,731 ± 812,062	8,985,211	18,946,021	3,012,605	13,998,252	0.477	0.07	430.1	3,753.2	1.019	0.19
v365	15	3:1,5:1,7:0	41·7·7	3	33.8	1,143,490	8,476,592 ± 840,677	7,823,724	19,958,933	3,357,060	14,527,031	0.506	0.56	426.7	3,663.9	0.994	−0.32
v5	21	3:1,5:0,7:1	5	2	0.6	518,996	8,203,783 ± 807,452	7,919,278	20,530,951	3,280,858	13,932,132	0.502	0.62	425.1	3,481.9	0.962	−1.01
v238	21	3:1,5:0,7:1	5·5·5	2	5.6	835,864	7,523,549 ± 907,473	6,600,507	25,150,503	2,429,717	14,129,846	0.615	1.14	426.5	3,202.6	0.882	−2.39
v340	21	3:1,5:0,7:1	41·5·13	6	18.7	604,319	8,621,001 ± 892,146	8,655,692	21,020,678	2,925,384	14,802,469	0.528	0.43	428.6	3,705.4	1.010	0.01
v4	35	3:0,5:1,7:1	3	1	1.9	679,102	8,044,880 ± 865,482	7,038,015	22,762,702	3,570,131	14,291,532	0.549	1.06	423.7	3,419.7	0.943	−1.31

v213	35	3:0,5:1,7:1	3·5·5	2	5.2	845,322	8,723,958 ± 912,587	7,927,579	18,847,252	2,790,159	14,593,579	0.534	0.13	424.4	3,715.1	1.023	0.24
v333	35	3:0,5:1,7:1	41·3·13	6	20.4	270,157	8,154,963 ± 822,229	7,745,654	24,197,893	3,206,687	14,400,466	0.514	0.71	425.7	3,498.3	0.956	−1.11
v17	45	3:2,5:1,7:0	3·7	3	2.3	667,336	8,638,410 ± 922,161	8,074,743	21,472,285	2,913,268	14,941,361	0.545	0.59	427.9	3,717.2	1.013	0.05
v136	45	3:2,5:1,7:0	3·59·7	3	50.2	124,133	8,237,870 ± 903,311	7,697,781	26,293,936	3,164,380	13,581,234	0.559	1.26	424.8	3,548.5	0.966	−0.83
v178	45	3:2,5:1,7:0	3·59·7	3	123.9	1,281,144	8,928,526 ± 824,596	9,097,302	19,696,424	4,196,682	14,679,506	0.471	0.44	425.7	3,921.1	1.047	0.76
v16	63	3:2,5:0,7:1	3·5	2	2.4	424,340	8,554,593 ± 820,873	8,592,585	19,989,231	2,662,157	13,538,865	0.490	0.15	427.0	3,649.5	1.003	−0.14
v135	63	3:2,5:0,7:1	3·59·5	2	50.0	153,958	8,904,979 ± 910,771	7,906,343	24,815,198	3,536,863	15,013,467	0.522	0.74	422.3	3,818.5	1.044	0.63
v177	63	3:2,5:0,7:1	3·59·5	2	123.9	440,510	9,129,770 ± 903,976	8,588,996	24,878,026	3,068,756	14,247,769	0.505	0.49	425.2	3,998.2	1.070	1.13
v0	105	3:1,5:1,7:1	root	—	0.0	1,208,721	8,719,464 ± 843,204	8,109,277	20,506,203	3,827,226	14,894,437	0.493	0.50	425.6	3,712.6	1.022	0.25
v8	105	3:1,5:1,7:1	23	11	4.7	1,243,917	8,200,940 ± 969,852	7,717,905	25,137,985	2,919,520	14,820,869	0.603	0.92	423.1	3,478.1	0.961	−0.85
v9	105	3:1,5:1,7:1	23	11	8.1	1,044,901	8,926,364 ± 864,264	8,299,106	21,584,874	3,391,351	14,313,271	0.494	0.43	424.4	3,796.5	1.046	0.72
v10	105	3:1,5:1,7:1	41	10	16.4	590,366	8,750,325 ± 888,182	7,837,885	21,864,753	3,510,993	14,819,376	0.518	0.73	424.3	3,733.8	1.026	0.30
v11	105	3:1,5:1,7:1	41	2	30.3	2,038,509	8,491,315 ± 963,999	7,777,558	32,970,081	3,307,435	13,855,630	0.579	1.42	425.5	3,654.4	0.995	−0.25
v257	105	3:1,5:1,7:1	7·7·13	6	5.5	318,590	7,896,295 ± 835,929	6,729,880	21,485,859	3,715,327	14,024,953	0.540	0.72	425.9	3,354.7	0.926	−1.70
v358	105	3:1,5:1,7:1	41·5·5	2	33.1	1,144,800	9,192,125 ± 968,973	8,017,046	23,142,860	4,022,689	16,864,134	0.538	0.90	425.6	3,946.6	1.077	1.19
v1	315	3:2,5:1,7:1	3	1	1.6	666,813	9,085,008 ± 931,133	8,867,891	22,017,266	2,585,750	14,544,636	0.523	0.47	423.3	3,829.0	1.065	1.01
v87	315	3:2,5:1,7:1	47·3	1	45.5	570,978	8,251,861 ± 816,730	7,978,601	22,240,447	3,327,101	13,118,212	0.505	0.87	423.5	3,549.5	0.967	−0.88
v158	315	3:2,5:1,7:1	3·23·13	6	10.0	798,175	8,447,891 ± 864,476	8,190,021	22,212,375	3,599,054	14,498,916	0.522	0.75	427.5	3,607.0	0.990	−0.38

P2: 20 generators of ⟨P⟩ per curve, the same secret, 10 paired seeds
each

curve	generators	solves	mean ± CI	min / max of generator means	sd of generator means (expected)	variance ratio	KW p	ANOVA p	max	z
v0	20	200	8,530,702 ± 625,427	5,609,911 / 12,180,728	1,951,735 (1,427,038)	2.060	0.021	0.008	2.70 (3.23)	−0.13 / 0.03 / −0.25 / −0.33
v158	20	200	9,253,752 ± 692,614	5,244,503 / 13,999,895	2,357,968 (1,580,338)	2.557	0.001	0.001	4.76 (3.23)	−0.07 / −0.15 / 0.01 / −0.02

All generator arms together: KW p = 0.000, variance ratio 2.336, flagged arms (family 40): [‘v158g12’, ‘v158g17’]; pooled mean 8,892,227 = 1.042 × √(πn/2); 400 solves, all correct.

Figures
Every figure draws recorded values only (`plot.py`). The standard volcano diagrams (`tools/volcano-diagram`) place each measured curve at its level in one ℓ-volcano with every walked edge drawn; the 3D heat maps stack the endomorphism-ring strata (rings, crater on top) and colour each measured curve by its mean operations or its mean end-to-end time (transport + solve).

3D isogeny-class heat map · p = 46339417407469 · mean group operations per solve
P2 · rings = endomorphism-ring strata (conductor f of End), 38 measured of 469 walked curves · expected √(πn/2) = 8,531,692

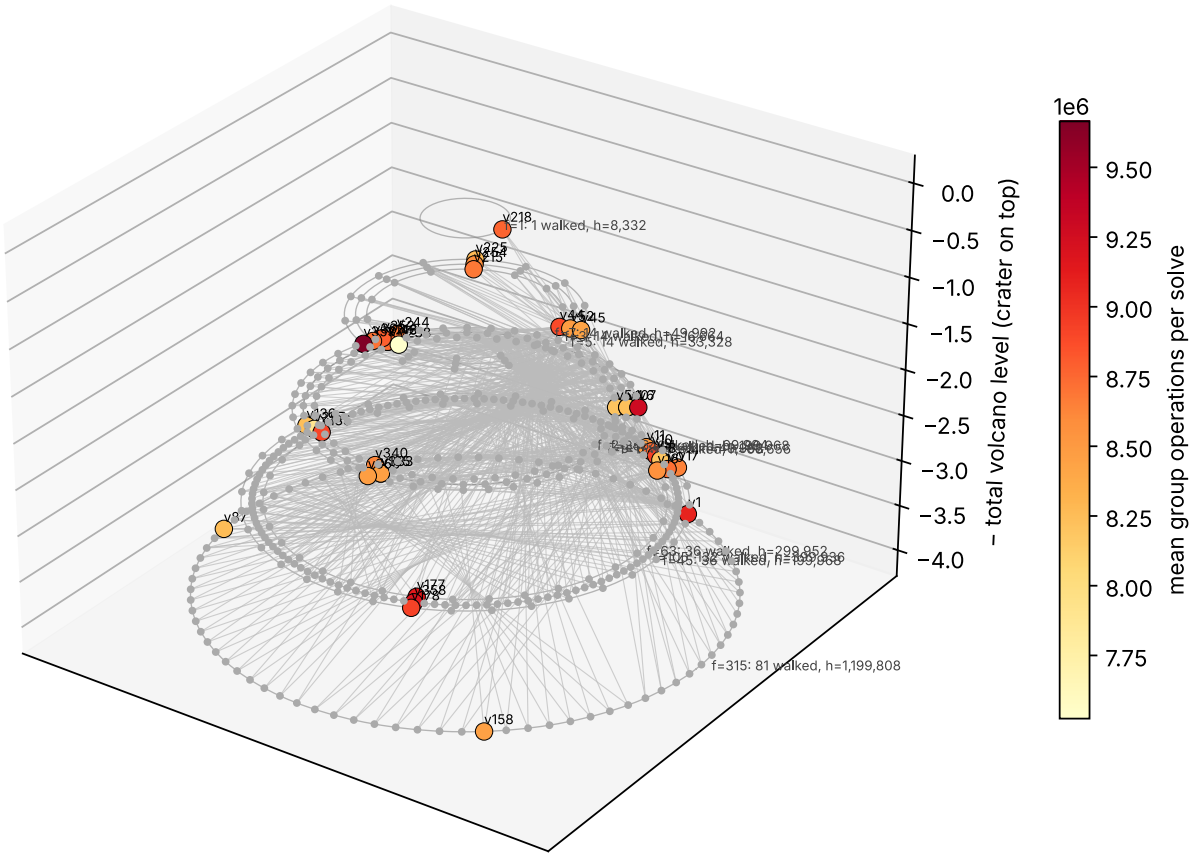


Figure 1: P2 3D class heat map, mean group operations per solve

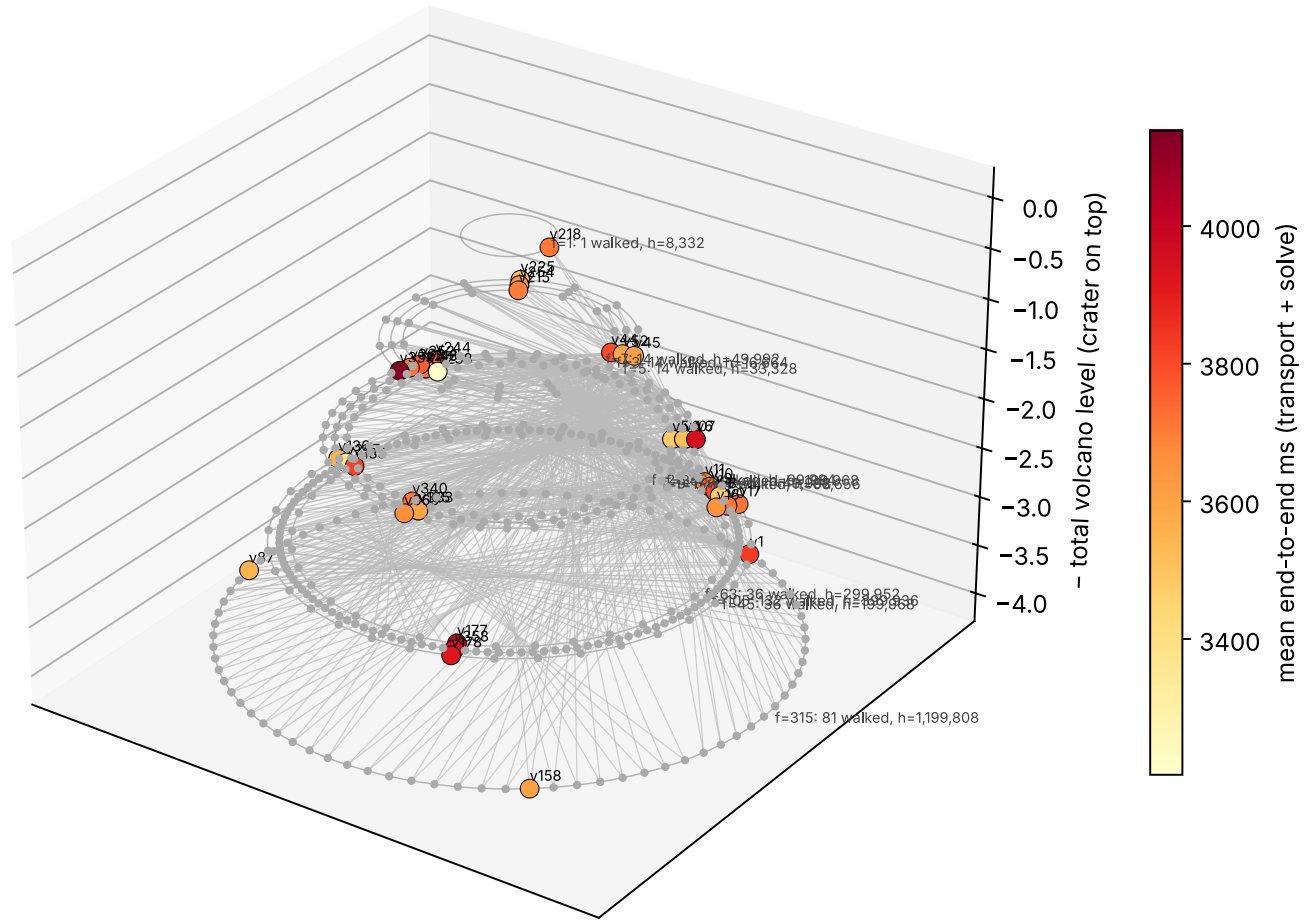


Figure 2: P2 3D class heat map, mean end-to-end ms (transport + solve)

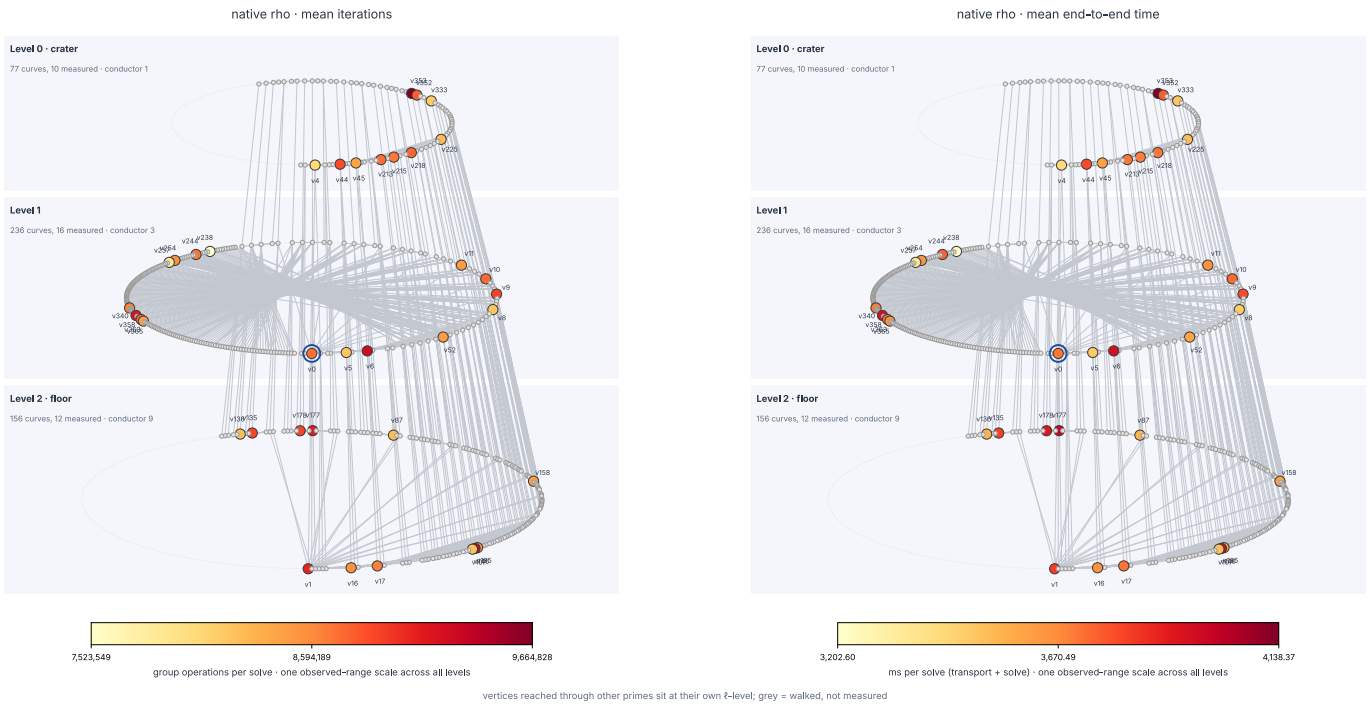


Figure 3: P2 volcano-ell3: standard volcano diagram

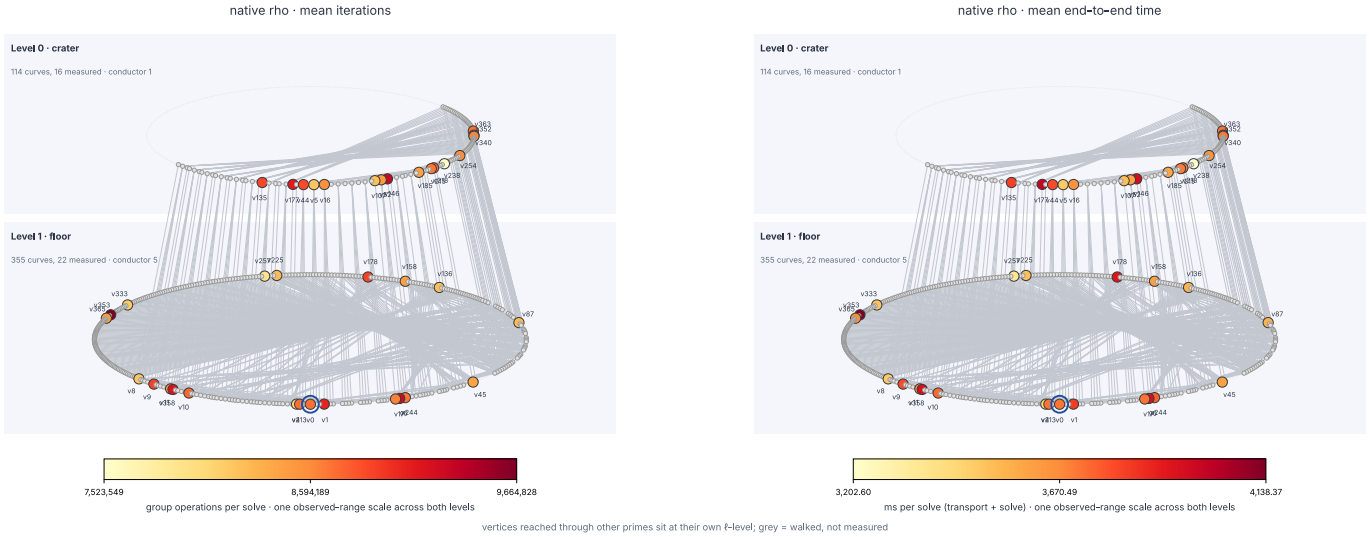


Figure 4: P2 volcano-ell5: standard volcano diagram

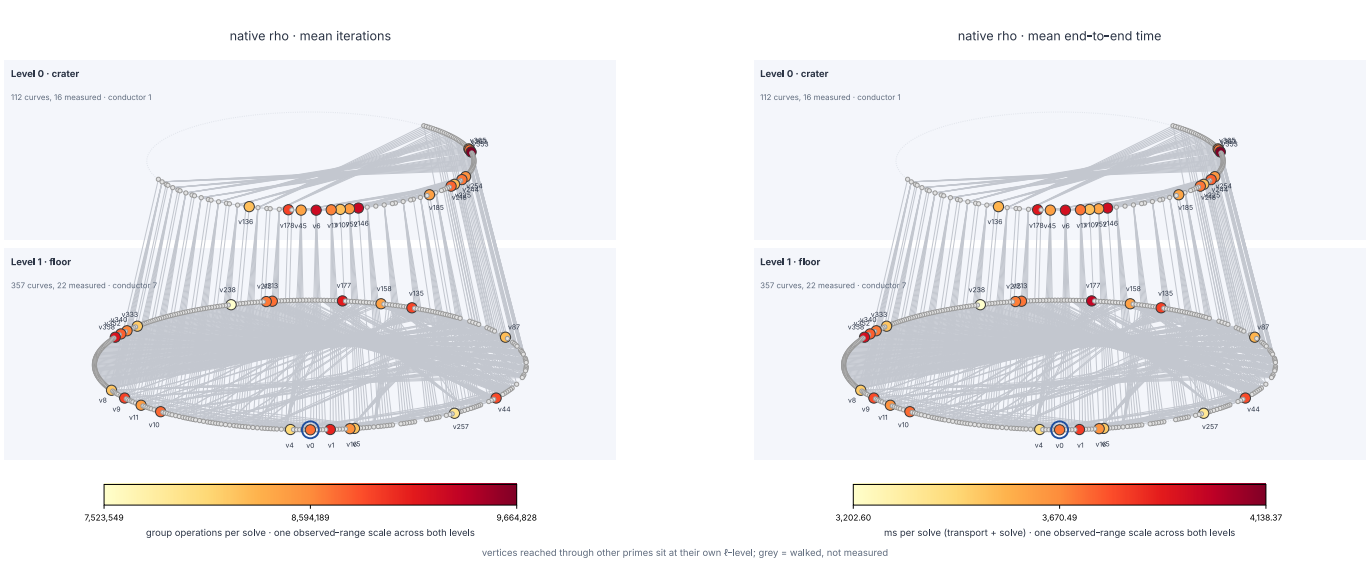


Figure 5: P2 volcano-ell7: standard volcano diagram

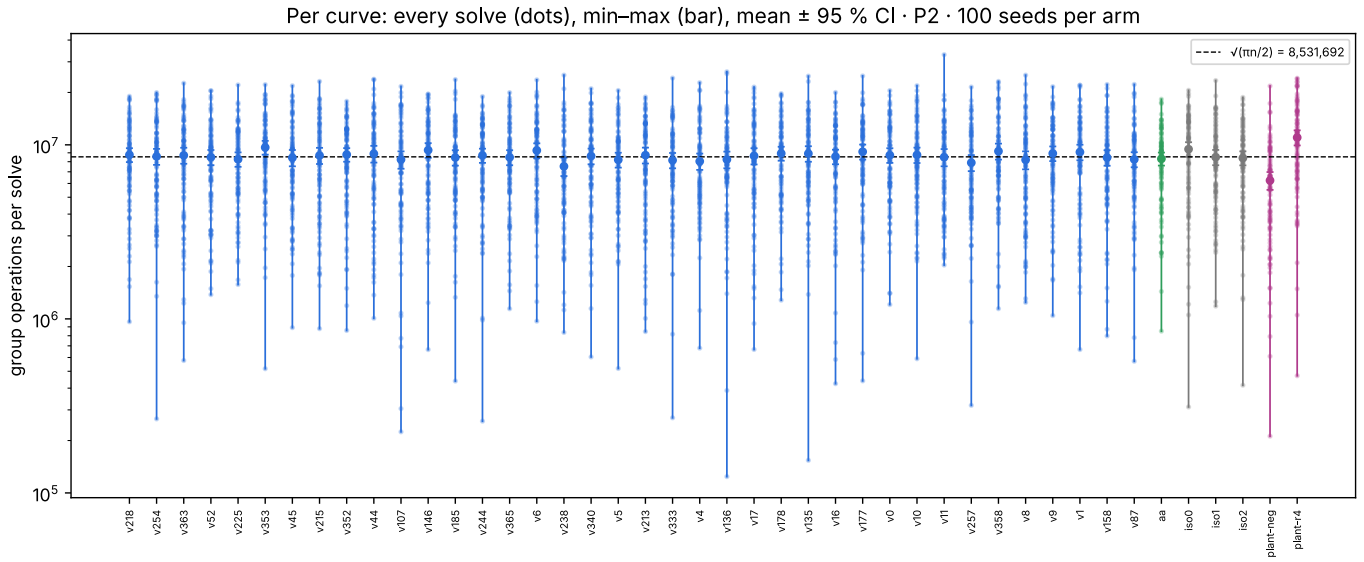


Figure 6: P2 every solve, min–max, mean ± CI per arm

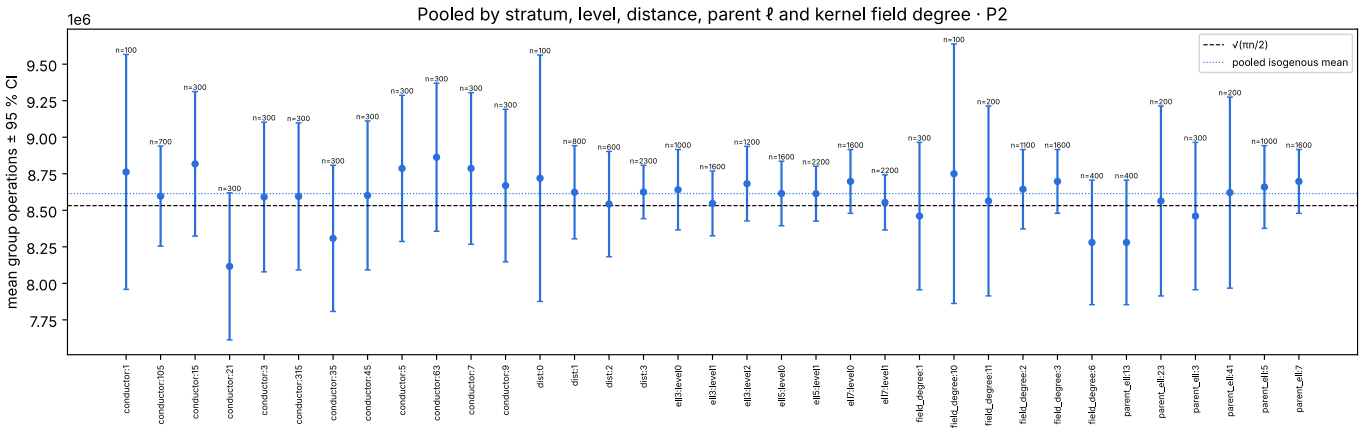


Figure 7: P2 pooled groups

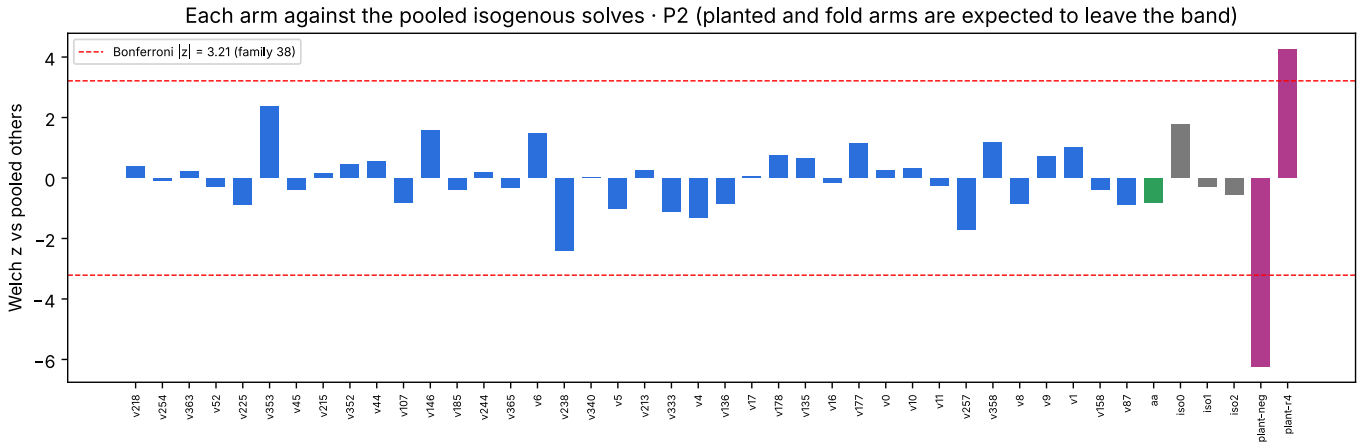


Figure 8: P2 each arm against the pooled isogenous solves

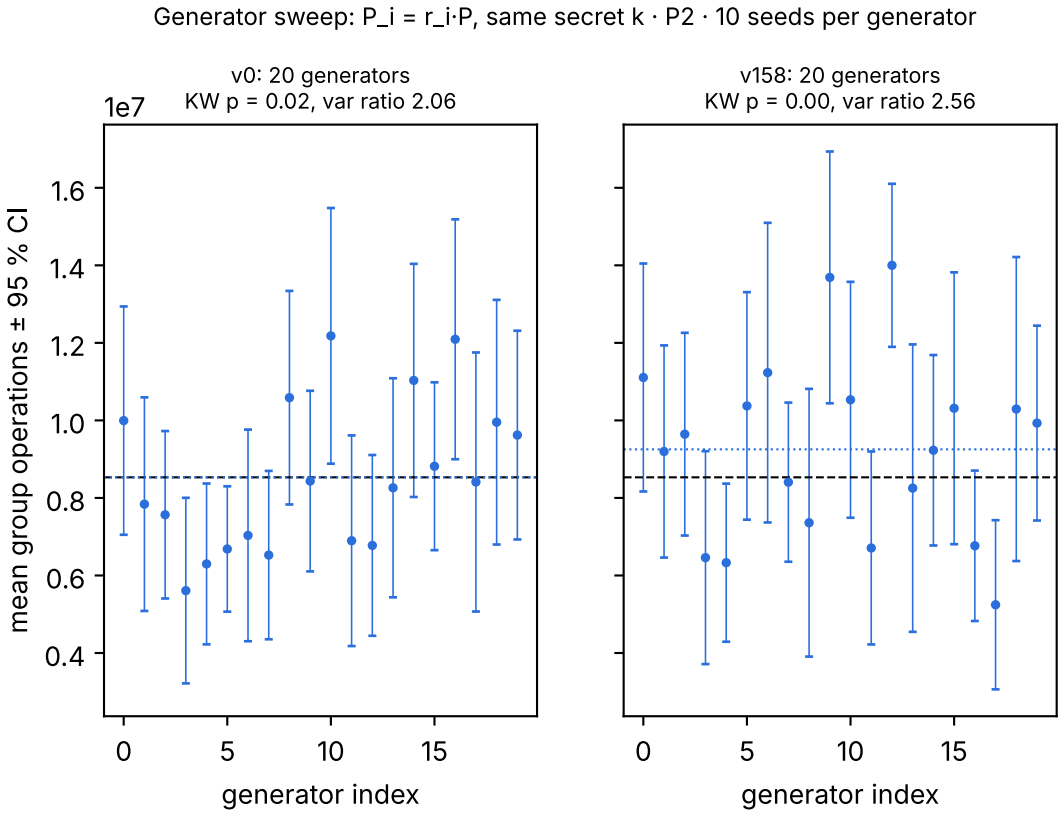


Figure 9: P2 generator sweep per curve

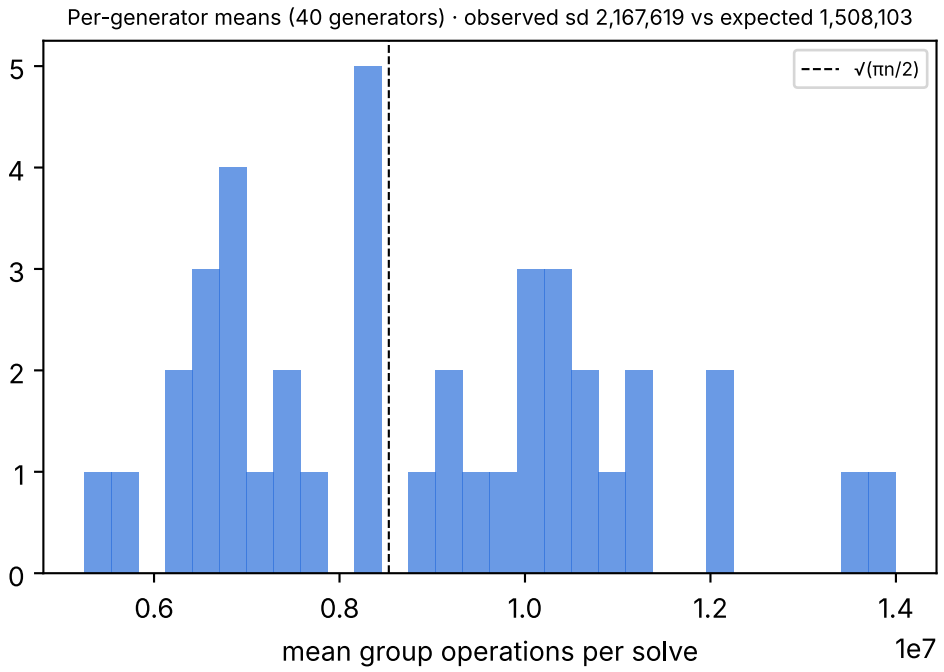


Figure 10: P2 distribution of generator means